# Alex Uamai

Mechatronics engineer building robotic machines for food production. Incoming M.S. Robotics, Johns Hopkins.

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## Profile

I take electromechanical hardware from a blank SolidWorks file to the factory floor, validated with FEA and real testing. I've also founded and scaled a startup to a 12-person team, and I'm now teaching robots in simulation.

## Experience

### Unifiller Systems
Dec 2025 – Present

Mechatronics Engineering Project Leader, R&D

- Own mechanical design of an industrial robotic food-automation platform, from concept to production support
- Design end-effectors, precision assemblies and fixtures in SolidWorks with GD&T and tolerance analysis
- Validate designs with FEA and first-principles analysis, then prototype, test and iterate on hardware
- Integrate motion, sensing, pneumatics and embedded electronics; drive DFM/DFA, BOMs and change control

### Grubchain
May – Dec 2025

Founder & CEO

- Founded a stablecoin payments company and grew the team from just me to 12
- Led product and engineering on a blockchain payment rail for merchants and consumers in Nigeria
- Represented the company on Channels Television (Sep 2025) and in a Vantech Journal op-ed (Jun 2025)

### Bethel Systems
Dec 2024 – Dec 2025

Technical & Business Analyst

- Turned early-stage client product ideas into engineering development plans, including support for patent applications

### VMAC Air
Sep – Dec 2021

Mechanical Engineering Intern

- Designed manufacturing fixtures and production test equipment in SolidWorks

### Ziva Dynamics
May – Aug 2021

Software Developer Intern

- Shipped production C++ and Python with a Qt/PySide2 interface, and contributed to Git-based automated testing

### Mulgrave School
Jan – May 2021

IT Intern

- Built a Raspberry Pi remote 3D-printing system and administered Windows Active Directory

## Projects

### Poundo Robot
Oct 2026 – Present

- Poundo yam, a West African staple and my favourite food, takes minutes of hard turning with a wooden stick to make. I'm building a robot arm that learns that motion from video, simulating an SO-101 in MuJoCo with Jacobian IK, PD control and contact sensing

### Two-wheel self-balancing robot

- Modeled dynamics and designed a state-space controller in MATLAB/Simulink, verified in Simscape 3D

### Lief, foot wearable for dystonia pain
Capstone

- Led controls: PWM vibration-motor firmware, the protoboard circuit and a desktop GUI for session tracking

### Embedded controllers

- STM32 tilt game controller (SPI, interrupts, USB); BLE board bring-up with J-Link; 8051 PWM motor control with encoders

## Education

### Johns Hopkins University
Starts Spring 2027

M.S., Robotics & Autonomous Systems Engineering

### Simon Fraser University
2018 – 2023

B.A.Sc., Mechatronic Systems Engineering

## Skills

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|---|---|
| **Robotics & sim** | MuJoCo, LeRobot, SO-101, inverse kinematics, PD control, PyTorch, industrial robotics, motion control |
| **Mechanical** | SolidWorks, GD&T, tolerance analysis, FEA, DFM/DFA, drawings & BOMs, rapid prototyping |
| **Electromech.** | Actuators, sensors, pneumatics, embedded systems, PCB integration, BLE, STM32 |
| **Software** | Python, C++, C, MATLAB/Simulink, Qt, Git, Linux, Jenkins, Jira, Linear, change management |